Unit 3.2 Printables Teacher Key: Age of the Earth

Standard: HS-ESS1-5. Evaluate evidence of the past and current movements of continental and oceanic crust and the theory of plate tectonics to explain the ages of crustal rocks. Targets are labeled ESS1-5.1 to ESS1-5.5 (priority) and ESS1-6.1 to ESS1-6.3 (supporting). Pacing follows the 26-27 workbook: 12 days across 10 lessons (Lessons 7 and 8 each take two days).

Data caveats

• Real data (ocean-floor ages, zircon, oldest rock, fossils, meteorites, polarity timescale, Hawaii) are rounded and cited on each item. Scientists revise ranges as new evidence arrives.

• Spreading rates, the age-versus-distance table (2.5 cm per year) and the North America ages are simplified for class (see item notes).

• The world map and continent cut-outs are simplified schematic outlines, not accurate coastlines. Do not use them to measure distance or area.

• Penny half-life results vary. Expect counts to halve on average, not exactly.

• The crater-count simulation counts depend on the slide images; the key gives the expected pattern.

• Q10 and Q12 on the CSA are the two written responses. They use the same claim, evidence, reasoning, and limit frames as Lesson 8.

How to use this key

Items are in lesson order and match the Printables index. Sample answers are model answers, not the only acceptable wording: accept any answer that uses the evidence correctly. Notebook warm-ups and exit tickets are listed under each lesson.

Lesson 1: Where Did the Old Ocean Floor Go?

Targets: Launch day: preview all unit targets (ESS1-5.1 to ESS1-5.5)

Pre-check (not graded)

Ungraded. Likely answers: 1) about 4.5 billion years; 2) answers vary; the oldest minerals are zircons about 4.4 billion years old (Western Australia) and the oldest rock is about 4.0 billion years old (Canada); 3) the ocean floor is younger, but many students will guess it is older; 4) tectonic plates of rock that move slowly; 5) radioactive (radiometric) dating, or by its layers and fossils. Compare at the end of the unit.

Hook data: ocean floor and continent ages

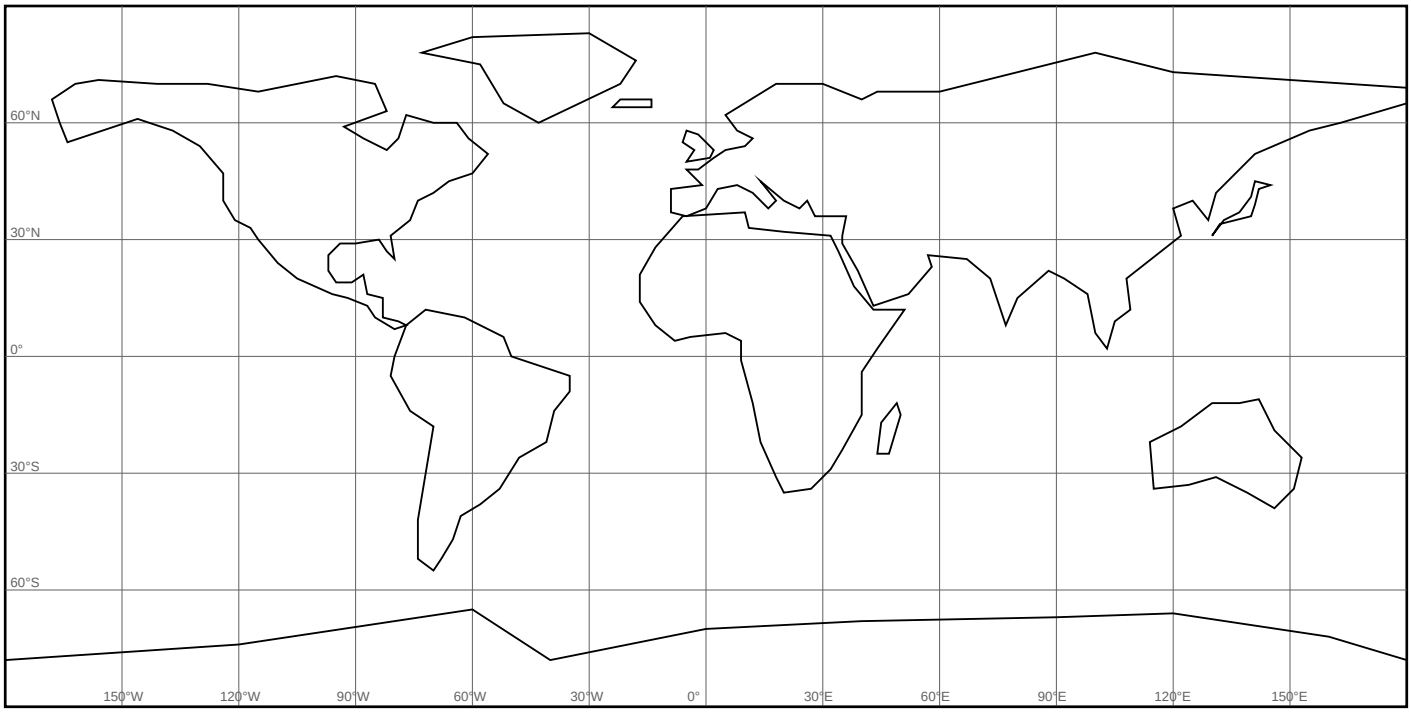
Notice: the youngest ocean floor is at the ridge and the oldest ocean floor is only about 180 million years old, while continental rock is billions of years old and fossils go back 3.5 billion years. Wonder: Why is the ocean floor so young? Where did the older ocean floor go? Student wording varies.

Unit CER page

Rows fill in over the unit. Example rows: Ocean floor ages increase away from the ridge (map, Lesson 3). Oldest ocean floor about 180 million years (data, Lesson 1). Magnetic stripes are mirror images across a ridge (model, Lesson 7). GPS and Hawaii give similar motion rates (data, Lesson 7).

World map and site list

Pattern: the points make belts (lines) that match the edges of the tectonic plates. Mount St. Helens, Fuji, Popocatépetl, Tohoku and Maule lie on the Pacific "Ring of Fire"; Hekla lies on the Mid-Atlantic Ridge; Erta Ale on the East African Rift; Haiti on the Caribbean plate boundary; Turkey on the Anatolian fault boundary. Kilauea (Hawai'i) is the site that does not fit: it is in the middle of the Pacific plate over a hot spot, not at a boundary.



Frayer card

Definitions: lithosphere (plate): Earth's rigid outer layer broken into moving plates. Plate boundary: where two plates meet. Mid-ocean ridge: where plates move apart and new ocean crust forms. Subduction: one plate sinks beneath another. Trench: deep groove where a plate sinks. Magnetic reversal: Earth's magnetic field flips direction; recorded in ocean-floor rock. Craton: a continent's ancient core. Impact crater: bowl-shaped hole made by a space rock.

Notebook sample answers

Warm-up: students guess; the ocean floor is younger. Exit ticket: the dots make belts (lines) that match plate edges. Vocabulary: see the Frayer card key.

Lesson 2: Three Ways Plates Meet

Targets: ESS1-5.1: I can describe Earth's outer layer as moving plates, name three kinds of plate boundaries, and describe where ocean crust is made and where it sinks back down.

Boundary table

Boundary	What the plates did	What formed	Real example
Divergent	Moved apart	Mid-ocean ridge; new ocean crust	Mid-Atlantic Ridge; East African Rift
Convergent	Moved together; one plate slid under the other	Trench, volcanoes, mountains	Andes (Nazca under South America); Japan trench
Transform	Slid past each other	Fault; earthquakes	San Andreas Fault, California

Ocean crust is made at a mid-ocean ridge (divergent boundary) and destroyed at a trench (convergent boundary, subduction).

Plate boundary word wall: 8 vocabulary cards

Cards already pair each word with its definition; for the matching game, cut each card in half. Check definitions against the Frayer card key (Lesson 1).

Notebook sample answers

Exit ticket: ocean crust is made at a mid-ocean ridge (divergent boundary) and sinks at a trench (convergent boundary).

Lesson 3: Reading Rock Ages

Targets: ESS1-5.2: I can read a rock-age map and describe the pattern in the ages.

Seafloor ages: table, color key and 13-box strip

Left and right sides are mirror images: youngest at the ridge (red), oldest farthest away (dark purple). As you move away from the ridge, the crust gets older. Strip colors: red, orange, yellow, green, blue, purple, dark purple, purple, blue, green, yellow, orange, red (symmetric about the ridge).

Graph grid: age versus distance

Points are (0,0), (500,20), (1000,40), (1500,60), (2000,80), (2500,100), (3000,120): a straight line. Rate = 500 km ÷ 20 million years = 25 km per million years = 2.5 cm per year. Fingernails grow about 3 to 4 cm per year (a few cm), similar to plate speed. At 3,500 km: 3,500 ÷ 25 = 140 million years.

North America ages: table and 5-box strip

West to east order: Pacific Coast Ranges (50 Ma, light yellow), Rocky Mountain region (150 Ma), Canadian Shield (2.7 Ga, dark brown), Grenville Province (1.1 Ga), Appalachians (400 Ma). The oldest crust is the Canadian Shield core. Age decreases as you move away from the core in both directions (tree-ring pattern). Note the strip order west to east is not oldest to youngest: young, medium, oldest, old, medium.

Notebook sample answers

Warm-up: the Brazil sample is older (farther from the ridge). Exit ticket: ocean crust gets older with distance from the ridge; youngest at the ridge.

Lesson 4: Clues to Early Earth

Targets: ESS1-6.1 (supporting): I can list the evidence scientists use to learn about early Earth.

45 cm Earth timeline: event table and three strips

Positions: oldest rock (4,500 – 4,000) ÷ 100 = 5 cm; oldest fossils 10 cm; oldest ocean floor 43 cm; today 45 cm. Only the last 2 cm of the 45 cm (200 million years) is covered by ocean floor that still exists. The oldest Earth rock is younger than Earth because early rock was melted, buried, or eroded and recycled by plate motion. There are no fossils that old in the ocean floor because the old ocean floor has been subducted and recycled.

Early Earth evidence log

Evidence list examples: zircon crystals (Western Australia, 4.4 billion years, solid crust existed); Acasta Gneiss (Canada, 4.0 billion years, oldest known rock); stromatolites and microfossils (Western Australia, 3.5 billion years, life existed); meteorites (4.56 billion years, solar system age); moon rocks (up to about 4.4 billion years, early crust). Earth rocks alone fail because plate tectonics, melting and erosion have destroyed the earliest rock; the oldest known rock is younger than Earth.

Notebook sample answers

Warm-up: students may say radioactive dating; accept reasoned ideas. Exit ticket: zircons, oldest rock, oldest fossils, meteorites, moon rocks.

Lesson 5: Meteorites and Moon Rocks

Targets: ESS1-6.2 (supporting): I can explain why meteorites and moon rocks tell Earth's age better than Earth rocks do.; ESS1-6.3 (supporting): I can use evidence from rocks, meteorites, and other planets to tell how Earth formed.

Crater counts: record sheet

Counts depend on the simulation slides: the Moon's total rises at each step (craters pile up); Earth's stays low (lava, erosion and plate motion clear craters). 1) The Moon: more craters, so older (more time to collect them). 2) Earth's craters are erased by weathering, erosion, volcanoes and moving plates; the Moon has none of these. 3) A surface with many craters is old; few craters means a young or resurfaced surface.

Half-life model: decay record

Real counts vary but roughly halve each shake: about 50, 25, 12, 6, 3, 2 (predicted column: 50, 25, 12.5, 6.25, 3.1, 1.6). 25% left = two half-lives: 2 × 700 million = 1,400 million years (1.4 billion years). Meteorites and moon rocks have been hardly changed since they formed, so they keep the original record; Earth's earliest rocks have been recycled by plate tectonics and weathering.

Notebook sample answers

Warm-up: Earth erases craters by erosion, lava, and moving plates. Exit ticket: meteorites and moon rocks have hardly changed since they formed, so they keep the early record; Earth rocks were recycled.

Lesson 6: Why the Ocean Is Young and the Continents Old

Targets: ESS1-5.3: I can explain why ocean crust is young and why continents are oldest at their ancient cores.

Guided notes: the life of ocean crust and how continents grow

Ridge; 120 million years; 5,000 km; 200 million years (about 180 to 200); ocean (denser) plate sinks; the edge is younger than the core.

Density lab table

Values depend on the blocks. Expected: wood density less than 1.0 (floats, like continental crust); clay density greater than 1.0 (sinks, like old dense ocean crust). The clay (ocean-plate) block dives under the wood (continent) block when pushed together; the water stands in for the mantle.

SketchNotes checklist

Check that the sketch includes every listed part and that the denser ocean plate is under the continental plate at the trench.

Notebook sample answers

Warm-up: it sinks at trenches (subduction) and is recycled. Exit ticket: ocean crust is young because it is made at ridges and recycled at trenches (denser plate sinks); continents are old because they are less dense and stay on the surface, growing from an ancient core.

Lesson 7: Evidence for Moving Crust

Targets: ESS1-5.4: I can judge how strong the evidence is that the crust has moved and is still moving.

Magnetic stripes: band table and paper strips

The two sides are mirror images: new crust forms at the ridge, records the field direction, and splits so each half moves away. This is evidence for seafloor spreading in the past (each band is an age). Strip colors: 1 blue, 2 red, 3 blue, 4 red from the outside in, mirrored on the other side.

Hawaii and GPS: measured motion

Kaua'i: 500 km ÷ 5 million years = 100 km per million years = 10 cm per year. Midway: 2,500 ÷ 28 = about 89 km per million years = about 9 cm per year. The rates (9 to 10 cm per year) are close to GPS (7 to 8 cm per year): independent methods (rock ages and satellites) agree that the Pacific plate moves several cm per year, so past and present motion are similar.

Pangaea puzzle: continent cutouts and fossil table

Best fit: South America nests into Africa; India, Antarctica and Australia sit near southern Africa. Mesosaurus could not swim across an ocean (fresh water) and heavy Glossopteris seeds could not blow across an ocean, so the continents must have been joined. Fossils show past positions but not current motion; GPS measures present-day motion.

Evaluate the evidence table

Evidence	Measured?	Past or present?	Independent?	Strength
Rock ages across a ridge	Yes (radiometric ages)	Past	Yes	4
Magnetic stripes	Yes (magnetometers)	Past	Yes	4
GPS and Hawaii	Yes (GPS and rock ages)	Present (and past with Hawaii)	Yes	4
Fossils and rocks	Partly (fossil location)	Past	Yes	3

Strongest for current motion: GPS. For past motion: rock ages and magnetic stripes (measured, mirror-image, repeatable). Two independent methods that agree are very unlikely to both be wrong. Accept reasoned ratings of 3 or 4 with justification.

Notebook sample answers

Exit ticket: strongest for past motion: rock ages across a ridge and magnetic stripes; for current motion: GPS.

Lesson 8: Evaluate the Theory, Tell Earth's Story

Targets: ESS1-5.5: I can judge how well plate tectonics explains the ages of rocks, using evidence.; ESS1-6.3 (supporting): I can use evidence from rocks, meteorites, and other planets to tell how Earth formed.

Three buckets: sort your evidence

Seafloor young: it forms at ridges and is subducted at trenches within about 200 million years (ages increase from the ridge; density). Continents old: continental crust is less dense and does not subduct; continents grow outward from ancient cores. Early record missing: plate recycling, melting and erosion destroyed the earliest rocks; only meteorites and moon rocks keep that record.

Part A sentence frames and data bank

Sample: Claim: Plate tectonics explains the ages very well because ocean crust is young where it forms and recycled at trenches, and continents survive. Ocean evidence: The seafloor is youngest at the ridge and gets older with distance; the oldest ocean floor is about 200 million years old. Continent evidence: Continental crust as old as 4.4 billion years exists. In North America the crust gets younger toward the edges. Reasoning: Because ocean plates are denser than continental crust, they sink at subduction zones, while continents stay at the surface. Limit: this evidence cannot show how the earliest crust formed (before 4 billion years); or: someone might say ocean floor is young because Earth was smaller then; the 4.4 billion-year zircons and meteorites do not fit that.

Star and step peer feedback

Student feedback varies. Look for specific stars (a number used, a clear reason) and one concrete step (add a number; explain why density matters; add a limit).

Earth's story: evidence cards and frames

Order oldest to newest: meteorites 4.56 Ga; moon highlands 4.4 Ga and Earth zircons 4.4 Ga (tie); oldest Earth rock 4.0 Ga; oldest fossils 3.5 Ga; oldest ocean floor 0.2 Ga. Story sample: About 4.56 billion years ago the solar system formed (meteorites). About 4.4 billion years ago solid crust existed (zircons, moon highlands). By 4.0 billion years rock survived in continents, and by 3.5 billion years life existed. Earth's earliest record is missing because plate recycling, melting and erosion destroyed it; the ocean floor is recycled in under 200 million years.

Notebook sample answers

Warm-up: students revise; look for plates, density, and evidence in the revised claim. Exit ticket: plate recycling, melting and erosion destroyed the early record; only meteorites and moon rocks keep it.

Lesson 9: Review Day

Targets: ESS1-5.1: I can describe Earth's outer layer as moving plates, name three kinds of plate boundaries, and describe where ocean crust is made and where it sinks back down.; ESS1-5.2: I can read a rock-age map and describe the pattern in the ages.; ESS1-5.3: I can explain why ocean crust is young and why continents are oldest at their ancient cores.; ESS1-5.4: I can judge how strong the evidence is that the crust has moved and is still moving.; ESS1-5.5: I can judge how well plate tectonics explains the ages of rocks, using evidence.

Self-rating chart and work plan

Student responses vary. Look for honest ratings, completed CFAs, and a specific plan that starts with the lowest target. ESS1-6 targets support ESS1-5.3 and 5.5 and are not on the CSA.

Review stations: five station cards

Station 1: ridge = divergent (new crust); trench = convergent (subduction); transform = sliding, earthquakes. Station 2: ages increase away from the ridge; (2,500 – 1,500) km ÷ (100 – 60) million years = 25 km per million years = 2.5 cm per year. Station 3: Ocean crust is made at ridges and sinks at trenches within about 200 million years; continental crust is less dense, does not sink, and keeps ancient cores. Station 4: GPS strongest for current motion; stripes and ages strongest for past motion. Station 5: see the Lesson 8 Part A key.

Notebook sample answers

Student responses vary.

Lesson 10: CSA

Targets: ESS1-5.1: I can describe Earth's outer layer as moving plates, name three kinds of plate boundaries, and describe where ocean crust is made and where it sinks back down.; ESS1-5.2: I can read a rock-age map and describe the pattern in the ages.; ESS1-5.3: I can explain why ocean crust is young and why continents are oldest at their ancient cores.; ESS1-5.4: I can judge how strong the evidence is that the crust has moved and is still moving.; ESS1-5.5: I can judge how well plate tectonics explains the ages of rocks, using evidence.

Notebook sample answers

Reflection: look for plates, density, subduction, and evidence (ages, stripes, GPS).

Matching table: evidence to the story it tells

Use this table to check Lesson 7 and Lesson 8 answers: match each line of evidence to what it shows.

Evidence	Kind	What it shows
Ocean-floor ages increase away from a ridge	Rock ages (map, graph)	Ocean crust is made at ridges and moves away
Mirror-image magnetic stripes on both sides of a ridge	Magnetic data (model)	Seafloor spreading in the past
Hawaii: island ages increase along the chain; GPS about 7 to 8 cm per year	Rock ages and GPS	The Pacific plate moves now and moved in the past
Same fossils on separated continents	Fossils	Continents were once joined
Oldest ocean floor is about 200 million years old	Rock ages	Ocean crust is recycled by subduction
Oldest continental rock is about 4.0 billion years old; zircons 4.4	Rock ages	Continents survive; early record is partial
Meteorites 4.56 billion years; moon highlands up to 4.4	Radiometric ages	Earth and the solar system formed about 4.5 billion years ago